Assignment 1

due: February 15, 2025

1. We set $T = 3, S_0 = 100, u = 1.2, d = 0.8, p_u = 0.7, p_d = 0.3$ and, for computational simplicity, $R = 0$. We now consider a particular contingent claim, namely a European put on the underlying stock. The date of expiration of the option is $T = 3$, and the strike price is chosen to be $K = 110$. Show that this particular claim can be replicated and compute the hedging portfolio at each node. (One needs to make a chart like Fig 2.10 in the textbook, which can also be found in page 48 of LN1 slides.)

2. Consider the multiperiod example in the text. Suppose that at time $t = 1$ the stock price has gone up to 120, and that the market price of the option turns out to be 50.0. Show explicitly how you can make an arbitrage profit.